6B ENTHALPY LEVEL DIAGRAMS

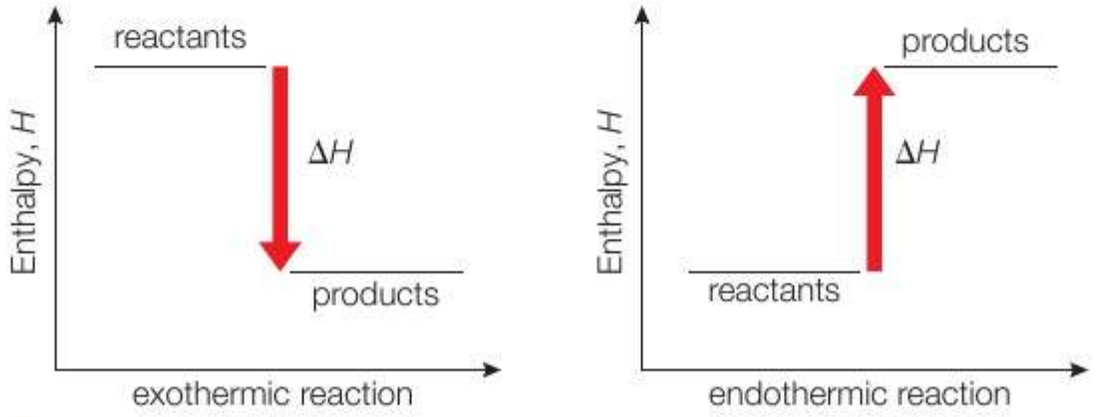
SPECIFICATION REFERENCE
6.2 6.3

LEARNING OBJECTIVES

- Know that, by convention, exothermic reactions have a negative enthalpy change and that endothermic reactions have a positive enthalpy change.
- Be able to construct and interpret enthalpy level diagrams showing an enthalpy change, including appropriate signs for exothermic and endothermic reactions.

HOW TO DRAW AND INTERPRET ENTHALPY LEVEL DIAGRAMS

In an exothermic reaction, the final enthalpy of the system is less than its initial enthalpy. The reverse is true for an endothermic process. This is shown in the two **enthalpy level diagrams** in **fig A**.



▲ **fig A** Enthalpy level diagrams for exothermic and endothermic reactions.

The change in enthalpy, ΔH, is given by:

$$\Delta H = H_{products} - H_{reactants}$$

For an exothermic reaction, $H_{reactants} > H_{products}$, so ΔH is negative.

For an endothermic reaction, $H_{reactants} < H_{products}$, so ΔH is positive.

LEARNING TIP

You must get two things right in labels for an enthalpy level diagram:

1. The + / − sign of ΔH must be correct
2. The direction of the arrow for ΔH must be correct.

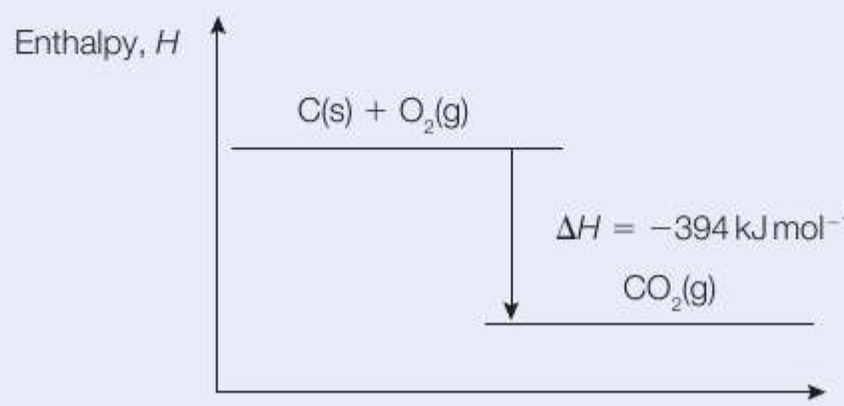
The arrow should always point towards the products, as the change is from the reactants to the products.

WORKED EXAMPLE 1

Draw an enthalpy level diagram for the following reaction:

$C(s) + O_2(g) \rightarrow CO_2(g)$ $\quad \Delta H = -394\ kJ\ mol^{-1}$

Answer



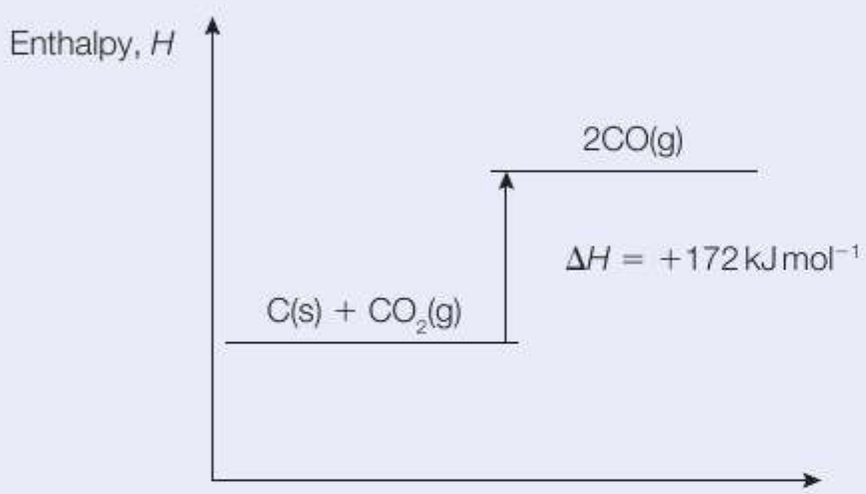
▲ **fig B** Enthalpy level diagram for the reaction of carbon with oxygen.

WORKED EXAMPLE 2

Draw an enthalpy level diagram for the following reaction:

$C(s) + CO_2(g) \rightarrow 2CO(g) \quad \Delta H = +172\,kJ\,mol^{-1}$

Answer



▲ **fig C** Enthalpy level diagram for the reaction of carbon with carbon dioxide.

Here are some points to remember when constructing enthalpy level diagrams.

- You only need to label the vertical axis. It is not necessary to label the horizontal axis in an enthalpy level diagram (if you did want to label it, you could use either 'Extent of reaction' or 'Progress of reaction'). It is, however, essential to label the horizontal axis in an enthalpy profile diagram (see **Topic 9** on reaction kinetics).
- The formulae for both reactants and products should be given, including their state symbols.
- The values for ΔH should be given, including the correct sign.
- It is not essential to show the activation energy in an enthalpy level diagram, but it should be shown in an enthalpy profile diagram (see **Topic 9** on reaction kinetics).

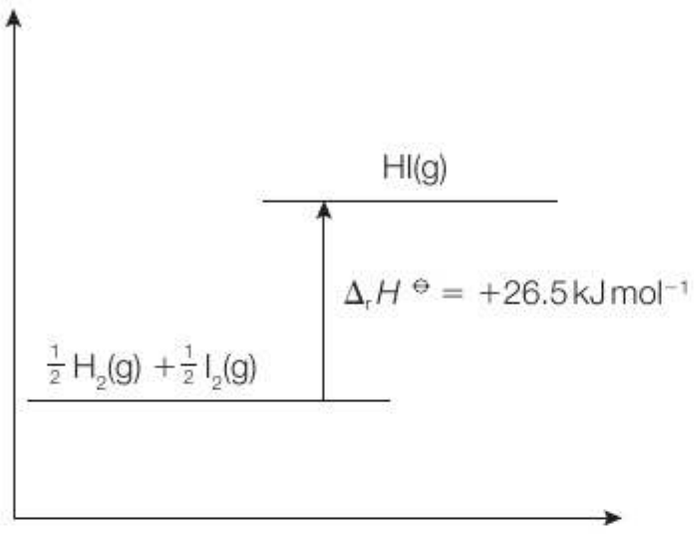
CHECKPOINT

1. Draw an enthalpy level diagram for the following reaction:

$CH_4(g) + 2O_2(g) \rightarrow CO_2(g) + 2H_2O(l) \quad \Delta H = -890\,kJ\,mol^{-1}$

2. (a) What information can be obtained from the following enthalpy level diagram?

SKILLS INTERPRETATION



(b) What would be the value of $\Delta_r H$ for this reaction?

$H_2(g) + I_2(g) \rightarrow 2HI(g)$

SUBJECT VOCABULARY

enthalpy level diagram a diagram that shows the relationship between the enthalpy of the reactants and the enthalpy of the products in a chemical reaction